

Diagnostic Study Framework

1. Purpose

- Place every learner at the level where they'll actually learn well — not too easy, not too hard.
- Base placement on demonstrated skill, not on age.
- Apply the exact same process across every track, so placement is consistent and predictable no matter which track a learner joins.

2. Scope

- Learners roughly 8–18 years old, currently in a K-12 school.
- Any board — CBSE, IGCSE, IB, or NGSS. Questions are written to be board-agnostic.
- Used only to qualify into Builder and Innovator. Explorer needs no test — it's the open starting point.
- A learner who can't decide on a track isn't forced to guess — they're placed in the Exploration track (a blend across all tracks) and skip this diagnostic altogether.

3. General Instructions

Aspect	Detail	Status
Delivery	Taken online through the test platform.	Confirmed
Focus / anti-cheating	The test window stays locked to the platform — no switching tabs. Closing the window auto-submits whatever's been answered.	Confirmed
Number of questions	25 to 30 per level test.	Confirmed
Total duration	45 minutes overall.	Confirmed
Time allocation	Up to 2 min/question; last 5 min set aside to review answers before submitting.	See note
Number of attempts	Unlimited — results aren't shown immediately, so retrying doesn't help a learner “peek” at answers.	Confirmed

Quick check: at a strict 2 min/question plus a 5-min recheck buffer, 45 minutes comfortably fits ~20 questions — a little under the 25–30 target. In practice most questions will take under the 2-minute cap, but it's worth confirming the final average once the question bank is locked.

4. Structure of the Framework

Every question in the bank is built from five pieces:

a) Skill Mapping

Each track has its own set of competencies aligned with standard skill categories. These categories apply across every track, and each map to the STREAMER pillars. Hence any learner who is confident in their skill competency will undergo this 360 diagnostic study to cover the required skills.

Skill Category	Streamer	What It Checks For
Conceptual Foundation	Science	The core ideas that explain how and why a design behaves the way it does.
Awareness of Materials	Engineering	Knowing which material fits which purpose — strength, weight, flexibility, and so on.
Equipment Handling	Engineering	Safe, correct use of the tools and equipment needed to build and maintain a project.
Circuit Basics	Technology	Understanding of circuits, wiring, and sensors needed to make a system function.
Creativity / Design Thinking	Arts	Generating original ideas and shaping them into something workable, not just copying a reference.
Data Analysis	Research	Reading results, spotting patterns, and drawing fair conclusions from tests.
Troubleshooting	Resilience	Staying systematic when something doesn't work, instead of guessing or giving up.
Decision Making	Entrepreneurship	Prioritising time, resources, and options sensibly, and adapting when circumstances change.

Mathematics is one of the eight STREAMER pillars that doesn't have its own skill row above — it currently runs in the form of Data Analysis (as numerical reasoning) and the Conceptual Foundation (as core formula).

b) Criteria

Every question carries a criteria tag — a short label naming the exact kind of thinking or skill it's testing. This keeps every question purposeful, and makes it easy to check that the bank covers a healthy mix of thinking types, not just fact recall. A question that doesn't fit a clear criteria tag doesn't belong in the bank.

Criteria Tag	What It Tests	Example
Reasoning	Working through cause and effect to reach a conclusion.	"Why does an object slow down as friction increases?"
Logical Thinking	Connecting a known rule to a new situation.	"If the switch is off, what happens to the rest of the circuit?"

Criteria Tag	What It Tests	Example
Pattern Recognition	Spotting a repeating trend across data or observations.	"Which of these three test results follows the clearest pattern?"
Concept Check	Confirming understanding of a core idea or definition.	"What does the term 'friction' mean?"
Material Selection	Choosing the right material for a given purpose.	"Which material is best suited for a lightweight, strong frame?"
Safety Check	Awareness of safe practice before or while handling tools or equipment.	"What should be checked before using a tool?"
Tool Handling	Correct and appropriate use of a specific tool.	"Which tool is most appropriate for tightening a small screw?"
Measurement	Accurately measuring or calculating a quantity.	"How would you measure the exact length of a component?"

This list isn't exhaustive — new tracks or new skill categories may call for additional tags. A single question can carry more than one tag if genuinely relevant, but most should have one clear, primary tag.

c) Question Type

All questions are multiple-choice, but five different formats are mixed across levels to test different kinds of thinking:

Type	What It Looks Like	Example
Textual	A plain question and answer in words — no visuals or numbers.	"Why does an object slow down as friction increases?"
Pictorial	Built around an image or diagram — identifying a part or reading a labelled picture.	Shown a labelled diagram, identify which part performs a given function.
Calculation-based	Uses formulas common across curricula, so it's fair regardless of board.	"If something travels 50m in 5 seconds, what is its speed?"
Scenario / Case study	A short real situation the learner has to reason through.	"A test result doesn't match what was expected — what's the first thing to check?"
Graph / Data interpretation	Reading a chart or data pattern and drawing a conclusion.	Given a chart of three test results, which one performed best, and why?

d) Options

- Keep language simple — plain words, unless the skill itself is technical vocabulary.
- Questions: maximum two lines.

- Answer options: under 8 words each — quick to scan, unambiguous.
- At Explorer level, keep the approach direct — one clear right answer, clearly wrong alternatives.
- At Builder level, use logical mapping — the learner connects a cause to an effect, or a concept to its correct application, instead of just recalling a fact.

5. Level Progression Flow

Every learner starts at Explorer by default. Skipping ahead requires clearing a test: the Explorer test to enter Builder, the Builder test to enter Innovator. Not clearing a test simply places the learner one level below where they aimed — there's always a clear next step.

